

Electrical & Sensor Fault Troubleshooting

Standard Operating Procedure

Doc ID	SOP-MNT-002
Version	2.0
Effective Date	2026-02-01
Owner	Maintenance Department
Scope	All machine types; sensors, limit switches, PLC I/O, and low-voltage control circuits

1. Purpose

Provides a structured diagnostic path for DT-04 sensor and limit-switch faults, the most frequent minor downtime code across the plant, to minimize repeat micro-stoppages.

2. Required PPE

- Insulated gloves
- Safety glasses
- Arc-flash rated clothing (for panel work >50V)

3. Tools & Materials

- Multimeter
- PLC diagnostic laptop
- Replacement proximity sensors (spec-matched)
- Wire ferrule kit

4. Procedure

- 1 Lockout-tagout the control circuit per Safety_Guidelines before opening any panel.
- 2 Check PLC fault log for the specific I/O address flagged under DT-04.
- 3 Visually inspect the sensor/switch for physical damage, misalignment, or debris.
- 4 Test sensor output voltage with multimeter against OEM spec; realign or clean if within tolerance.
- 5 If sensor is out of tolerance or non-responsive, replace with spec-matched unit and re-teach the PLC if required.
- 6 Cycle the affected station manually 5 times to confirm the fault is cleared before resuming production.
- 7 Log the fault code, root cause, and corrective action in the CMMS.

5. Associated Downtime Code

Code	Description	Severity	Typical Duration
DT-04	Sensor / Limit Switch Fault	Minor	8 min

■ De-energize and verify zero voltage with a rated multimeter before touching any terminal inside a control panel.